

Overnight Basal Plan

Andrew · Omnipod + Loop · Dexcom G7 · dorm since Aug 21 · **data through Aug 23, 2026**

Personal working notes from Nightscout data. One change at a time; watch, then decide. Not medical advice.

WHERE THINGS STAND

Overall control	mean ~110, GMI ~5.9%
Time below 54 (last 2 wks)	1.8 – 3.1%
Recent overnight nadirs	40 · 43 · 50
Bolus insulin	33.9 U/day (was 43.3)
Scheduled basal	38.05 U/day
Basal Loop actually gives	~24.5 U/day

Average glucose is fine — this is a **hypoglycemia exposure** problem, not a control problem. Two things changed in August: marching band added activity, and he moved to a dorm where he eats late.

The schedule has drifted **56% above what Loop actually delivers**. Loop suspends 27–38% of the overnight hours holding it down — and the scheduled rate is what the pod falls back to if Loop stops.

THE PLAN

- 1

Late-correction rules — start tonight, no settings change
The single biggest lever on the 2am crashes, and it costs nothing to try. See §3.
- 2

Overnight basal reduction — 22:30 / 00:00 / 02:00
The settings change. See §2. Give it 5–7 nights before judging.
- 3

Daytime basal — deferred
09:30 and 16:00 are both 1.80 U/hr and also well above delivery (Loop suspends 24–62%), but changing them risks daytime highs that get over-corrected. Revisit once the ~9am-class routine has a few weeks of data.

1 · WHY OVERNIGHT — SCHEDULED VS DELIVERED

hour	sched	loop gives	susp.	%<54
22:00	2.40	1.32	36%	0.3
23:00	2.20	1.28	38%	0.5
00:00	1.45	0.92	31%	0.5
01:00	1.45	0.88	33%	0.2
02:00	1.30	0.72	37%	1.3
03:00	1.30	0.90	27%	0.0
04:00	1.30	1.00	16%	0.0
05–08	1.35	1.01–1.11	10–15%	0.0

Green rows are the part Loop accepts — near the programmed rate, barely suspended, no lows. Leave 04:30–09:30 alone; cutting it is the change most likely to create morning highs. Red rows are where schedule and reality diverge.

2 · THE CHANGE

22:30	2.20 → 1.70
00:00	1.45 → 1.20
02:00	1.30 → 1.00
04:30	1.35 → unchanged
09:30+	daytime unchanged

Daily total	38.05 → 36.05 U (–5.3%)
All of it between	22:30 and 04:30
vs Loop's actual	+58–81% → +30–39%

This lowers real delivery, not just the fallback: during the ~65% of the night Loop isn't suspended, a lower baseline means less insulin. It doesn't limit Loop's ability to respond upward — it can still temp above schedule. **If it under-delivers, 1.55 / 1.10 / 0.90 still sits ~20–25% above what Loop gives.**

3 · LATE CORRECTIONS — THE BIGGEST LEVER, AND IT WORKS TONIGHT

overnight bolus source	per night
22:00–02:00 · manual	3.74 U
22:00–02:00 · Loop auto	1.57 U
03:00–05:00 · Loop auto	1.32 U
BG when manual doses given	share
Under 120	30%
120–150	30%
150–180	32%
180+	8%

- ×

No manual correction under ~150 after 10pm.
60% of manual doses go in below 150 — that's dosing ahead of a rise, not treating a high. Late-meal peaks are modest anyway (median 159; only 3 of 12 passed 180).
- ×

No Rage mode at night. 1.2× insulin with an 80 target is a daytime tool. Same for fake carbs — both recruit subcutaneous insulin whose 6-hour tail is still active at 3–4am.
- ✓

Afrezza if a late correction is genuinely needed.
~90-minute action, gone before 2am. Matches fast carby food far better than subcutaneous, which outlasts the meal by hours.
- ✓

Substitute, don't add. Afrezza *instead of* the subcutaneous correction. At BG 110 on a rise, even Afrezza can overshoot.

The cycle in the data: manual insulin 30–60 min after the meal, rescue carbs at 120–180 min, more insulin at 180–210 min, repeating for five hours. Insulin given at 22:00–23:00 is still working at 03:00–04:00. On Aug 22 a Rage correction at 01:00 with BG **127** was followed by **43** ninety minutes later.

4 · WHAT TO WATCH AFTER EACH CHANGE

Run: `python3 check_response.py` — it already excludes sensor warmup and flags G7 compression.

Working: overnight (00:00–05:00) time below 54 drops toward <1%, no nadir under 60, and morning (07:00–09:00) mean stays under about 140.

Too far: morning mean climbs past ~160, or he starts correcting in the morning. Back the overnight rates up halfway.

Not enough: severe lows persist at 00:00–05:00 after 5–7 clean nights → go to the second step (1.55 / 1.10 / 0.90).

Ignore: single bad nights, and anything in the first 24h of a new sensor. Judge on 5–7 nights, excluding warmup.

5 · DELIBERATELY NOT CHANGING

setting	current	why not
Correction target	100 flat, 24h	Not changing targets by preference. Note it does mean Loop keeps correcting toward 100 overnight — 81% of its automatic overnight insulin goes in below 150.
ISF	53	The other lever on Loop's aggressiveness (53 → 58 would soften every correction). Unvalidatable while Afrezza is unlogged — hold.
Carb ratio	6	Late-meal dosing already runs slightly weaker than this (implied 6.6) and late-meal peaks are modest. Not the problem.
Daytime basal	09:30 & 16:00 at 1.80	Both well above delivery, but cutting risks daytime highs → over-correction. Deferred, not dismissed.
04:30–09:30 basal	1.35	The one block Loop accepts. Produces no lows. Leave it.

6 · WHAT THIS DATA CAN'T SEE

Afrezza is invisible. Not logged, so every insulin total here is a floor and any correction analysis undercounts.

Fake carbs inflate carb totals. Small entries used to prompt Loop to correct aren't food.

Prebolus timing is invisible — carbs and insulin are logged at the same timestamp, so the log shows dose time, not eat time.

The activity sample isn't the semester. Early-start days (n=7) ran 4.0% below 54 vs 0.9% on late-start days (n=52) — but those were band camp plus one Saturday event, not a normal ~9am-class week. Anything activity-shaped needs re-deriving in a few weeks. **The scheduled-vs-delivered gap does not depend on this** — it holds on every day type.

G7 compression lows are frequent overnight — brief untreated dips that rebound on their own. Excluded from the numbers here; they would otherwise roughly double the apparent low rate.